

SHETH L.U.J AND SIR M.V COLLEGE
PRACTICAL :- 2
SUB :- Artificial Intelligence

CODE :- # AIM: Implement IDDFS (Iterative Deepening Depth-First Search)
Graph: Uttarakhand to Mumbai

```
dict_gn = dict(
Dehradun=dict(Haridwar=55, Mussoorie=35),
Haridwar=dict(Dehradun=55, Roorkee=30),
Mussoorie=dict(Dehradun=35, Rishikesh=75),
Roorkee=dict(Haridwar=30, Rishikesh=70),
Rishikesh=dict(Mussoorie=75, Roorkee=70, Delhi=240),
Delhi=dict(Rishikesh=240, Jaipur=280),
Jaipur=dict(Delhi=280, Ahmedabad=660),
Ahmedabad=dict(Jaipur=660, Surat=270),
Surat=dict(Ahmedabad=270, Vapi=120),
Vapi=dict(Surat=120, Virar=170),
Virar=dict(Vapi=170, Borivali=35),
Borivali=dict(Virar=35, Kandivali=5),
Kandivali=dict(Borivali=5, Malad=4),
Malad=dict(Kandivali=4, Goregaon=4),
Goregaon=dict(Malad=4, Andheri=5),
Andheri=dict(Goregaon=5, Bandra=6),
Bandra=dict(Andheri=6, Dadar=7),
Dadar=dict(Bandra=7, Mumbai=8),
Mumbai=dict(Dadar=8)
)

start = "Dehradun"
goal = "Mumbai"
result = ""

def DLS(city, visitedstack, startlimit, endlimit):
    global result
    found = 0

    result = result + city + " "
    visitedstack.append(city)
```

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```
if city == goal:
    return 1

if startlimit == endlimit:
    return 0

for eachcity in dict_gn[city]:
    if eachcity not in visitedstack:
        found = DLS(eachcity, visitedstack, startlimit + 1, endlimit)
        if found:
            return found

return 0

def IDDFS(city, visitedstack, endlimit):
    global result

    for i in range(endlimit + 1):
        print("Searching at Limit:", i)

        found = DLS(city, visitedstack, 0, i)

        if found:
            print("Found")
            break
        else:
            print("Not Found!")
            print(result)
            print("-----")
            result = ""
            visitedstack = []

def main():
    visitedstack = []

    IDDFS(start, visitedstack, 20)

    print("\nIDDFS Traversal from", start, "to", goal, "is:")
```

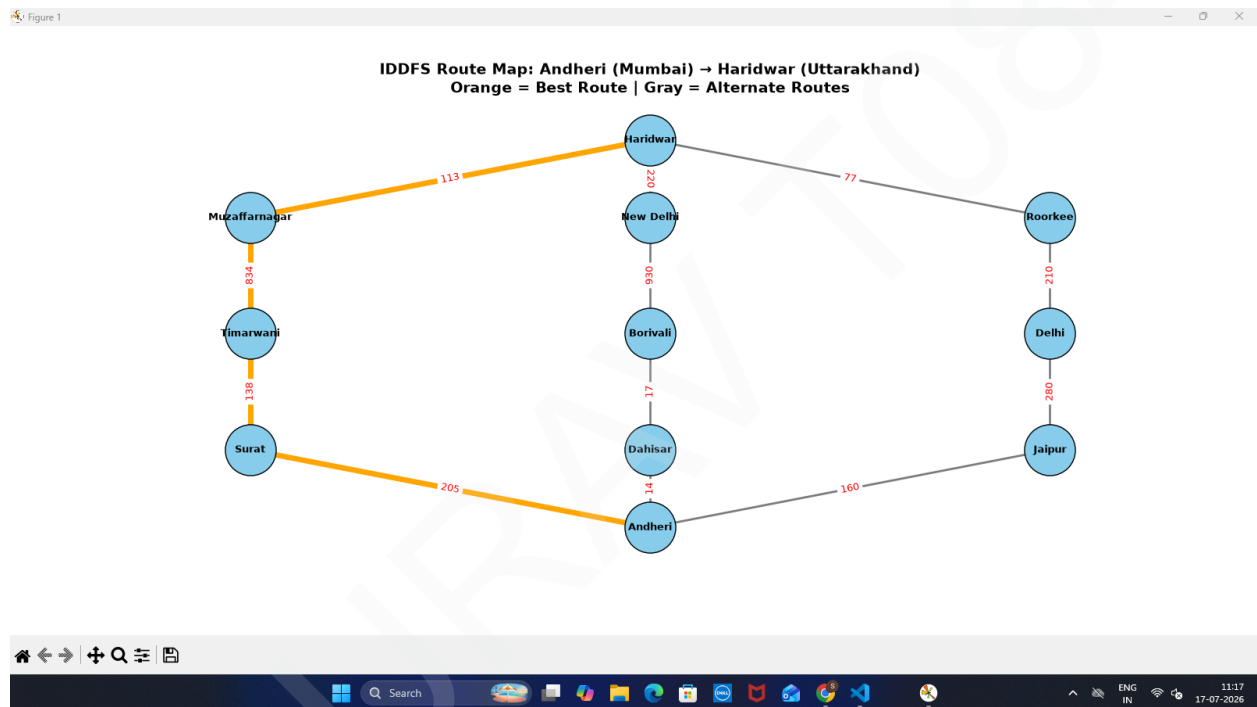
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```
print(result)

main()
```

OUTPUT :-

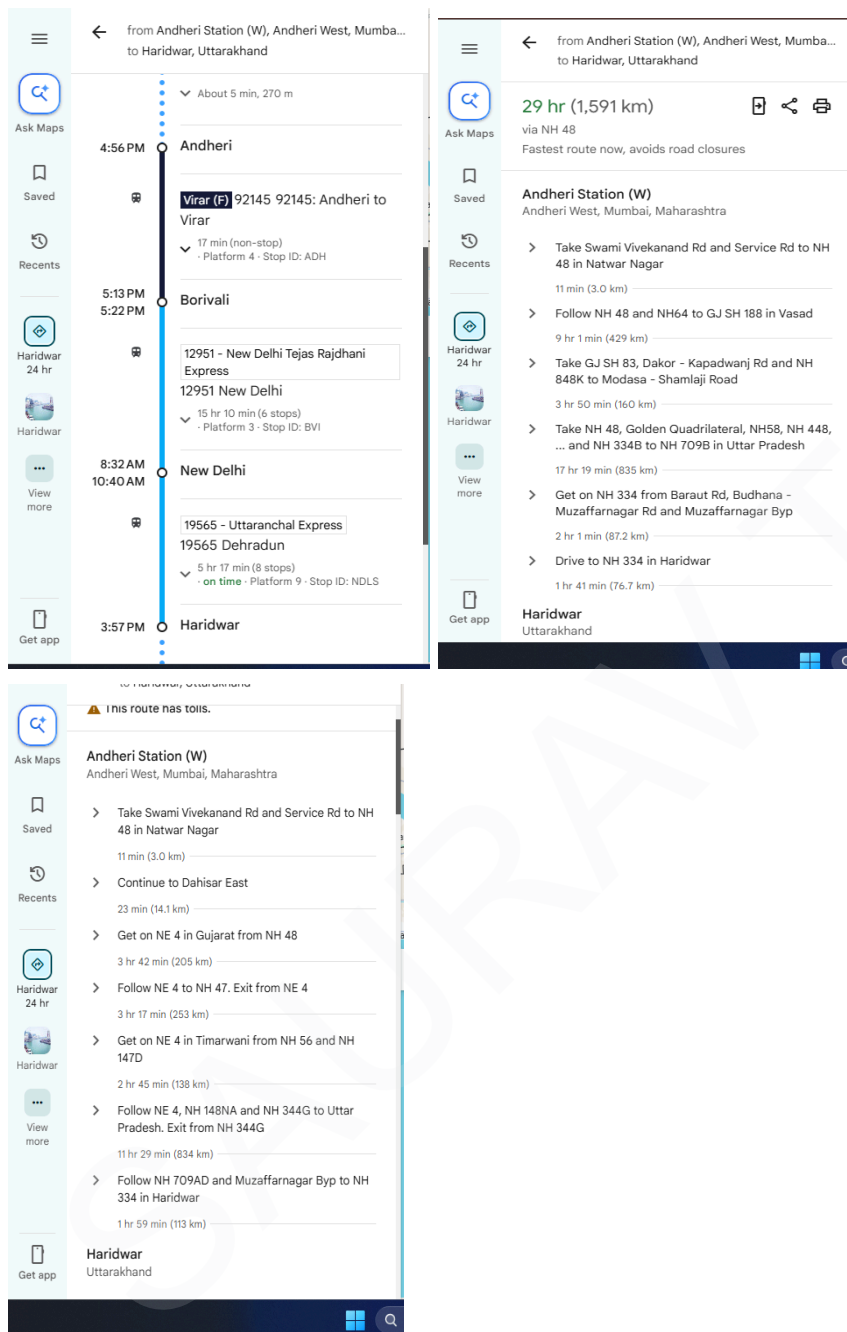


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